

Exercise 1: Create Table (DDL) Create a table named students with: id INT PRIMARY KEY, full_name VARCHAR(50), gpa DECIMAL(3,1), enrollment_date DATE.

Exercise 1.1: Additional Table Create a table named courses with: course_id INT PRIMARY KEY, course_name VARCHAR(100) NOT NULL, credit_hours INT CHECK(credit_hours BETWEEN 1 AND 5).

Exercise 1.2: Relationship Table Create a table enrollments with: student_id INT, course_id INT, grade VARCHAR(2), PRIMARY KEY(student_id, course_id), FOREIGN KEY(student_id) REFERENCES students(id), FOREIGN KEY(course_id) REFERENCES courses(course_id).

Exercise 2: Insert Data Insert two students: (1, 'Ali Hassan', 3.8, '2023-09-01') and (2, 'Sarah Ahmed', 2.9, '2023-09-05').

Exercise 2.1: Insert Data into courses Insert: (10, 'Database Systems', 3) and (11, 'Machine Learning', 4).

Exercise 2.2: Insert Enrollment Records Insert enrollments linking students to courses.

Exercise 3: Simple UPDATE Update Sarah Ahmed's GPA to 3.2 using her id.

Exercise 4: Conditional UPDATE Update enrollment_date to '2024-01-01' only for students enrolled on '2023-09-01'.

Exercise 5: Primary Key Violation Test Try inserting another student with id = 1. What happens and why?

Exercise 6: CHECK Constraint Test Try inserting a course with credit_hours = 10. What error should appear?

Exercise 7: Foreign Key Violation Test Try inserting enrollment with student_id = 99. Why does it fail?